Lab 1 warm-up

Hello World test

```
print("Hello World")
```

```
Hello World
```

Part B: Python warm-up

1. Variables and arithmetic

```
# Calculate the volume of a rectangular estuary
# Formula: length × width × depth

length = 5000
width = 800
depth = 3.5

volume = length * width * depth

print(f"Estuary volume: {volume} m^3")
```

```
Estuary volume: 14000000.0 m^3
```

2. Lists and loops

```
# Simulate 10 days of exponential population growth

population = 100
rate = 0.05
history = [population]

for day in range(10):
    population = population + rate * population
    history.append(round(population, 1))

print(history)
```

```
[100, 105.0, 110.2, 115.8, 121.6, 127.6, 134.0, 140.7, 147.7, 155.1, 162.9]
```

3. Functions

```
# Convert temperature from Celsius to Fahrenheit

def celsius_to_fahrenheit(c):
    return c * 9/5 + 32

# Test the function

for temp in [0, 15, 25, 37]:
    print(f"{temp}°C = {celsius_to_fahrenheit(temp)}°F")
```

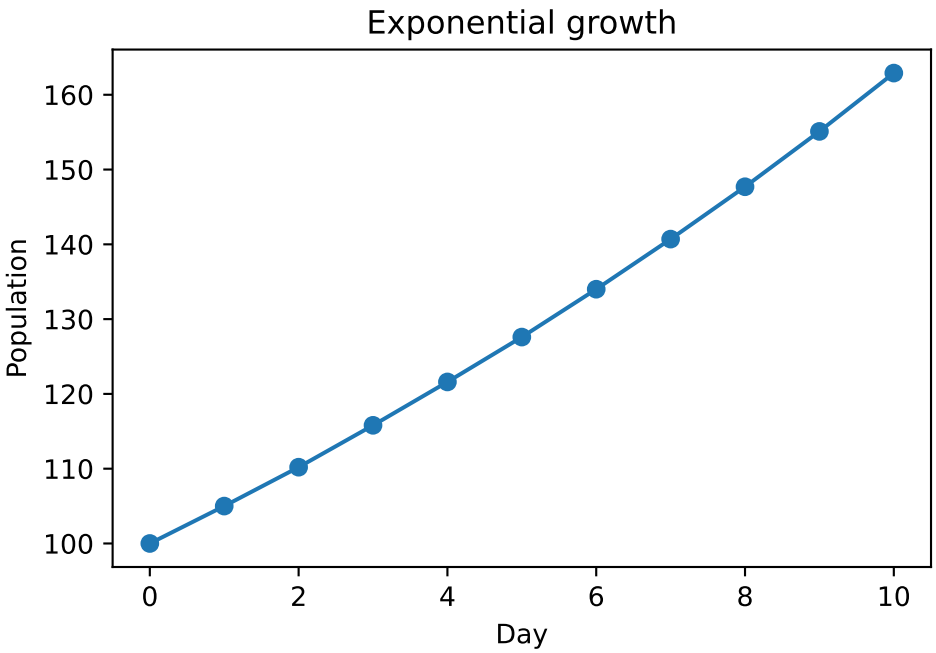
```
0°C = 32.0°F
15°C = 59.0°F
25°C = 77.0°F
37°C = 98.6°F
```

4. Plotting

```
import matplotlib.pyplot as plt

days = list(range(len(history)))
```

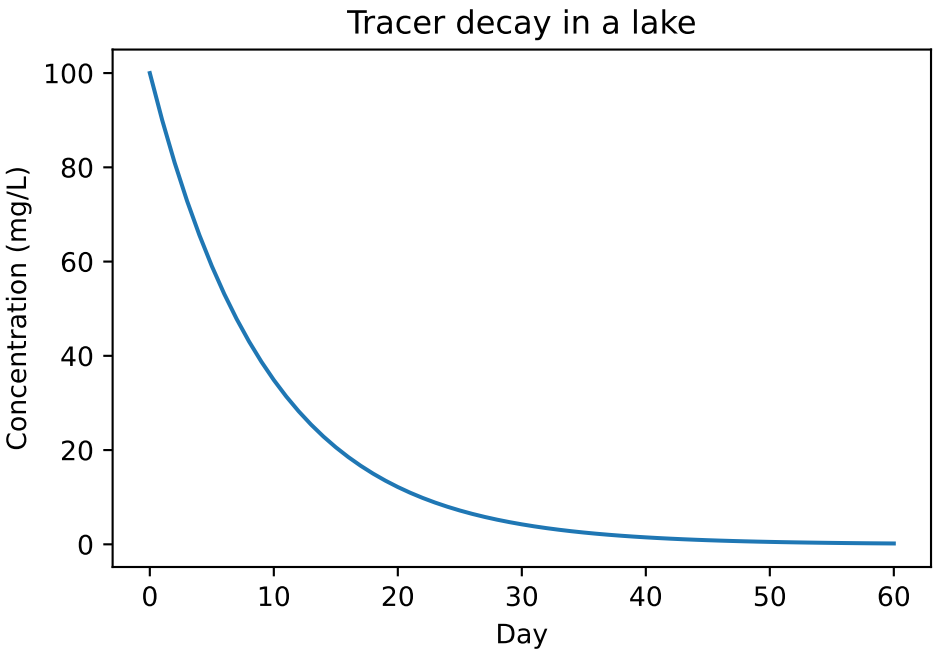
```
plt.plot(days, history, "o-")
plt.xlabel("Day")
plt.ylabel("Population")
plt.title("Exponential growth")
plt.show()
```



Part C: Modelling preview

```
import matplotlib.pyplot as plt

concentration = 100.0
decay_rate = 0.1
dt = 1.0
tracer_history = [concentration]

for day in range(60):
    concentration = concentration - decay_rate * concentration * dt
    tracer_history.append(concentration)
```

```
plt.plot(tracer_history)
plt.xlabel("Day")
plt.ylabel("Concentration (mg/L)")
plt.title("Tracer decay in a lake")
plt.show()
```